

Healthcare Infusion Analytics Pipeline

Problem Statement

THE PROBLEM

Infusion-center operational data is distributed across multiple source tables for patients, appointments, order series, medication administration, locations, and treatment details. In its raw form, the data does not provide a single trusted view of infusion activity, making routine operational questions difficult to answer consistently and creating risk of duplicate records, inconsistent joins, and unclear treatment definitions.

Business Need

- Create a reliable analytical foundation that converts raw operational records into a reusable infusion-level dataset.
- Provide consistent metrics for patient volume, successfully administered treatments, drug/location utilization, weekly operations, and data quality.
- Make the logic transparent and reproducible so results can be validated and extended into BI or operational reporting.

Project Objective

Build an end-to-end local analytics pipeline using Python, Pandas, DuckDB, and SQL that ingests the source workbook, validates table relationships, constructs a curated **fact_infusions** table, executes business queries, and exports reporting-ready CSV outputs.

Key Questions to Answer

Patient Volume

How many unique patients received infusions by month?

Treatment Completion

How many treatments were successfully administered in Q1 2026?

Drug & Location

Which drug/location combinations have the highest administered infusion counts?

Monthly Leaders

Which drug/location combination leads each month?

Weekly Operations

How do infusion volume, unique patients, PIV attempts, and WoW change by location/drug?

Data Quality

Where are missing fields, invalid PIV values, or status/medication inconsistencies?

Core Analytical Assumptions

- Appointments are treated as the infusion event grain; meds_administered = 'Yes' defines a successfully administered treatment.
- Because administered medication volume is not present, administered infusion count is used as the available proxy for volume.
- The latest order-series record per patient is used for drug mapping to reduce appointment duplication risk.

Expected Outcome

A validated, reproducible analytics workflow that produces a trusted infusion fact table and reporting-ready outputs for operational monitoring, business analysis, and future dashboard development.